AYESHA ARSHAD

Research Analyst | Data Analyst | AI Enthusiast

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PROFESSIONAL SUMMARY

Results-driven BS Statistics (Data Science) student at COMSATS University Islamabad with hands-on experience in statistical analysis, data automation, and academic research. Proficient in Python, R, SPSS, and Power BI, with a track record of building AI-powered multi-agent systems and end-to-end data pipelines that turn raw data into actionable insight. Recognized as a Kaggle Notebook Expert (Global Rank #882). Completed a data-focused internship at the Pakistan Bureau of Statistics and a Deloitte Australia Data Analytics job simulation. Seeking a Research Analyst, Data Analyst, or AI Engineering Intern role to apply rigorous analytical skills and deliver high-quality, evidence-based results.

SKILLS

- **Programming Languages:** Python, R, SQL
- **Agentic AI & LLMs:** Multi-Agent Systems, Prompt Engineering, LLM Orchestration, AI Workflow Design, Explainable AI (XAI), Responsible AI, AI Safety, Natural Language Processing (NLP)
- **Web Scraping & Automation:** Selenium, Playwright, BeautifulSoup, Pandas, CSV Data Pipelines
- **Statistical Tools:** R, SPSS
- **Data Visualization:** Power BI, ggplot2, Tableau
- **Productivity & Office:** Microsoft Word, Excel, PowerPoint
- **Version Control:** Git, GitHub
- **Analytical Competencies:** Exploratory Data Analysis, Research Design, Statistical Modelling, Problem Solving
- **Academic & Professional Writing:** Research Proposals, Business Plans, Structured Reports
- **Soft Skills:** Attention to Detail, Time Management, Teamwork, Collaboration

EXPERIENCE

Pakistan Bureau of Statistics (PBS) Intern **Aug 2025 – Sep 2025**

Lahore, Pakistan

- Collected, cleaned, and organized statistical datasets to support departmental reporting and analysis.
- Prepared and structured survey and statistical data for downstream analysis, improving dataset readiness for reporting.
- Used Microsoft Excel and statistical tools to manage, validate, and organize official datasets.
- Maintained data accuracy and consistency across structured records used in statistical work.

EDUCATION

COMSATS University Islamabad BS Statistics (Data Science) **2023 – Present**

Lahore, Pakistan

- Pursuing a rigorous curriculum covering statistical modelling, data science, and research methodology.
- Completed applied academic projects in civic research and entrepreneurship, producing professional-grade documentation.

Commerce College of Sheikhupura Intermediate in Commerce (I.Com) **2020**

- Achieved Grade A, demonstrating strong academic performance across commerce and quantitative subjects.

MC School, Sheikhupura Matriculation (Science) **2018**

- Completed secondary education with a science stream focus.

PROJECTS

Silent Witness: Multi-Agent AI System for Manipulation & Dark Pattern Detection

Technologies: Agentic AI, LLMs, Prompt Engineering, Multi-Agent Systems, Python, Streamlit, GitHub

- Designed and built an AI-powered multi-agent system that detects manipulative language and dark patterns in emails, chats, and marketing messages.
- Architected a modular 4-agent pipeline (Detector, Explainer, Responder, and Guardian) to enable collaborative analysis and decision-making.

- Engineered prompt workflows to flag manipulation tactics such as fake urgency, emotional blackmail, authority bias, guilt-tripping, and misleading social proof, with transparent explanations for each flag.
- Implemented an end-to-end pipeline that classifies suspicious content, explains detected tactics, and recommends safer, user-protective responses.
- Prioritized explainability and responsible AI design, producing interpretable outputs instead of black-box predictions.
- Collaborated with a cross-functional team on system architecture and agent orchestration.

Web Scraping & Data Automation Project

Python, Selenium, Playwright, BeautifulSoup, Pandas

- Built a Python-based web scraping system to collect business data, including names, ratings, reviews, addresses, phone numbers, websites, emails, categories, opening hours, and Google Maps URLs.
- Automated data extraction, cleaning, validation, and CSV export workflows to streamline dataset delivery.
- Added filtering logic to retain only records with usable contact information (email, website, or social media presence), improving output data quality.

Ecommerce EDA: Exploratory Data Analysis

Python

- Conducted end-to-end exploratory data analysis on an e-commerce dataset to uncover sales patterns, customer behavior, and product trends.
- Applied data cleaning, statistical summaries, and visualization techniques using Python.

Gradify: GPA & CGPA Calculator

Python

- Developed a Python application to automate GPA and CGPA calculations, enabling students to track academic performance efficiently.
- Applied software engineering principles to deliver clean, user-friendly input/output handling.

Summary Stats & Summary Stats 2: Statistical Analysis Tools

Python

- Built Python scripts to compute descriptive statistics and summary metrics on structured datasets.
- Extended the toolkit with EDA capabilities for deeper data insights (Summary Stats 2).

Scientific Calculator

Python

- Designed and implemented a scientific calculator supporting trigonometric, logarithmic, and exponential functions.
- Demonstrated strong Python programming fundamentals and modular code design.

Basic Calculator

Python

- Built a clean, functional arithmetic calculator in Python as a foundational programming project.

Civics & Ideology Research Project

Academic

- Conducted systematic research on civic and ideological principles, synthesizing findings from multiple academic sources.
- Authored a professionally formatted report in Microsoft Word, demonstrating strong academic writing and document structuring skills.

Entrepreneurship: Business Plan Development

Academic

- Designed and delivered a full business plan covering market analysis, financial forecasts, and multi-channel marketing strategies.
- Produced polished professional documentation in Microsoft Word, applying business writing conventions.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Deloitte Australia Data Analytics Job Simulation, Forage. Completed a job simulation covering data analysis and forensic technology; built a data dashboard in Tableau; used Excel to classify data and draw business conclusions (2026).
- Power BI Workshop Certification: data visualization principles, interactive dashboard design, and reporting best practices.
- Data Analytics with R Certification: data manipulation, statistical analysis, and visualization using R, including tidyverse and ggplot2 workflows.
- Microsoft Office Specialist (MOS) Certification: expert-level competency in Microsoft Word, Excel, and PowerPoint.
- Power BI Advanced Course (In Progress): DAX, data modelling, and enterprise-level reporting solutions.

- Introduction to APIs in Python. DataCamp
- Intermediate Git. DataCamp
- Intermediate GitHub Concepts. DataCamp
- ETL and ELT in Python. DataCamp
- Software Engineering Principles in Python. DataCamp
- Associate Data Engineer in SQL. DataCamp
- Introduction to Apache Airflow in Python. DataCamp

LANGUAGES

English (Professional) | Urdu (Native) | Punjabi (Native)